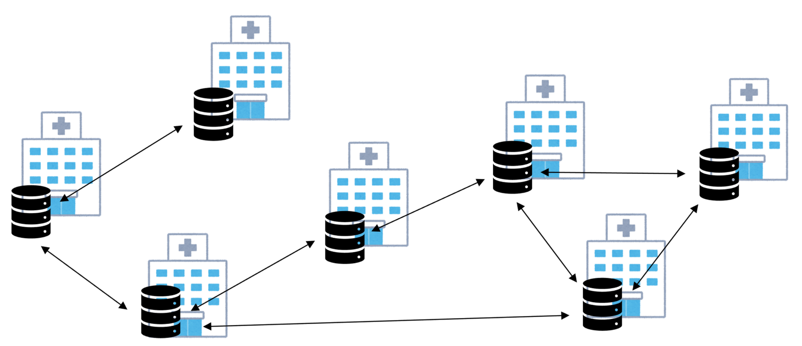
Scalable Decentralized Learning with Teleportation

Yuki Takezawa[1,2], Sebastian U. Stich[3]

[1]Kyoto University, [2]OIST, [3]CISPA

Background

- **Decentralized learning**
  - Decentralized learning can preserve privacy because it does not need to aggregate all training data into one server.



  - Let the number of nodes be n and the loss function of node i be $f_i(x)$, decentralized learning is formulated as follows:

$$\min_{x \in \mathbb{R}^d} \left[f(x) \coloneqq \frac{1}{n} \sum_{i=1}^{n} f_i(x) \right] \quad (1)$$

- **Decentralized SGD**

$$x_i^{(t+1)} = \sum_{j=1}^{n} W_{ij} \left(x_j^{(t)} - \eta \nabla F_j(x_j^{(t)}; \xi_j^{(t)}) \right). \quad (2)$$

Assumption 1 (L-smoothness)

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \quad (\forall x, y \in \mathbb{R}^d) \quad (3)$$

Assumption 2 (Stochastic Gradient Noise)

$$\mathbb{E}\|\nabla F_i(x; \xi_i) - \nabla f_i(x)\|^2 \leq \sigma^2 \quad (\forall x \in \mathbb{R}^d) \quad (4)$$

Assumption 3 (Heterogeneity)

$$\frac{1}{n} \sum_{i=1}^{n} \|\nabla f_i(x) - \nabla f(x)\|^2 \leq \zeta^2 \quad (\forall x \in \mathbb{R}^d) \quad (5)$$

Assumption 4 (Spectral Gap)

There exists $p_n \in (0, 1]$ that satisfies

$$\|XW - \bar{X}\|_F^2 \leq (1 - p_n) \|X - \bar{X}\|_F^2, \quad (6)$$

for any $X \in \mathbb{R}^{d \times n}$ where $\bar{X} \coloneqq \frac{1}{n} X \mathbf{1}\mathbf{1}^\top$.

Generally, p_n approaches zero as n increases.

Proposition (Decentralized SGD)

Suppose that Assumptions 1-4 hold. Then, there exists a stepsize η so that $\frac{1}{T+1} \sum_{t=0}^{T} \mathbb{E}\|\nabla f(\bar{x}^{(t)})\|^2$ is bounded from above by

$$\mathcal{O}\left(\sqrt{\frac{L r_0 \sigma^2}{nT}} + \left(\frac{L^2 r_0^2 (p_n \sigma^2 + \zeta^2)(1 - p_n)}{T^2 p_n^2} \right)^{\frac{1}{3}} + \frac{L r_0}{T p_n} \right), \quad (7)$$

where $\bar{x}^{(t)} \coloneqq \frac{1}{T} \sum_{i=1}^{n} x_i^{(t)}$ and $r_0 \coloneqq f(\bar{x}^{(0)}) - f^\star$.

It requires more iterations as n increases since p_n approaches zero.

Contribution

Q: Can we develop a decentralized learning method whose convergence rate does not degrade as the number of nodes increases?

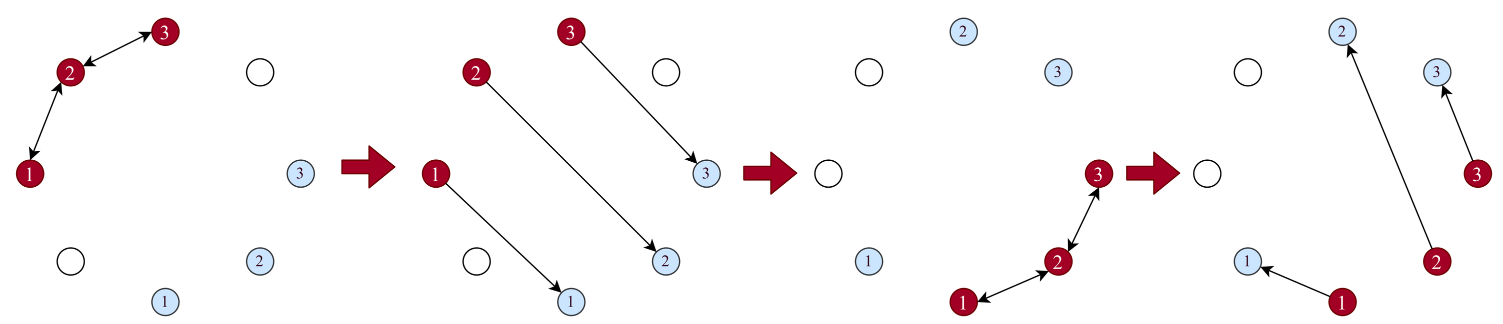
A: Yes. TELEPORTATION can completely eliminate the degradation of the convergence rate caused by an increase in the number of nodes.

Proposed Method

- **Key idea**
  - In many scenarios, the topology merely represents a preferred communication pattern, and connecting to arbitrary nodes is still possible, e.g., in a data center.
  - In these cases, we can simulate Decentralized SGD with a small number of nodes even if the total number of nodes is extremely large.

Algorithm 1 TELEPORTATION

Require: n: total number of nodes, k: the number of active nodes.
  for $t = 0, 1, \ldots, T$ **do**
    Sample active nodes $V_{\text{active}}^{(t)}$ from all nodes without replacement.
    The active nodes $V_{\text{active}}^{(t)}$ fetch the parameter of previous active nodes $V_{\text{active}}^{(t-1)}$.
    The active nodes $V_{\text{active}}^{(t)}$ initialize their parameters with the fetched parameters.
    Connect active nodes $V_{\text{active}}^{(t)}$ by sparse topology, e.g., ring.
    Run Decentralized SGD once on $V_{\text{active}}^{(t)}$.
  end for



- **Convergence analysis**
  - The convergence rate can be consistently improved as n increases.

Theorem 1 (Teleportation)

Suppose that Assumptions 1-3 hold. Let $\{\{x_i^{(t)}\}_{i=1}^n\}_{t=0}^T$ be the parameters generated by *TELEPORTATION*, and Suppose that the active nodes are connected by a ring. Then, if we set the number of active nodes k as follows:

$$k = \max\left\{1, \min\left\{ \left\lceil \left(\frac{T(\sigma^2 + \zeta^2)}{L r_0} \right)^{\frac{1}{7}} \right\rceil, n \right\}\right\},$$

there exists η such that $\frac{1}{T+1} \sum_{t=0}^{T} \mathbb{E}\|\nabla f(\bar{x}_{\text{active}}^{(t)})\|^2$ is bounded from above by

$$\mathcal{O}\left(\sqrt{\frac{L r_0 \sigma^2}{nT}} + \left(\frac{L r_0 (\sigma^2 + \zeta^2)^{\frac{3}{4}}}{T} \right)^{\frac{4}{7}} + \left(\frac{L r_0 (\sigma^2 + \zeta^2)^{\frac{2}{5}}}{T} \right)^{\frac{5}{7}} + \frac{L r_0}{T} \right),$$

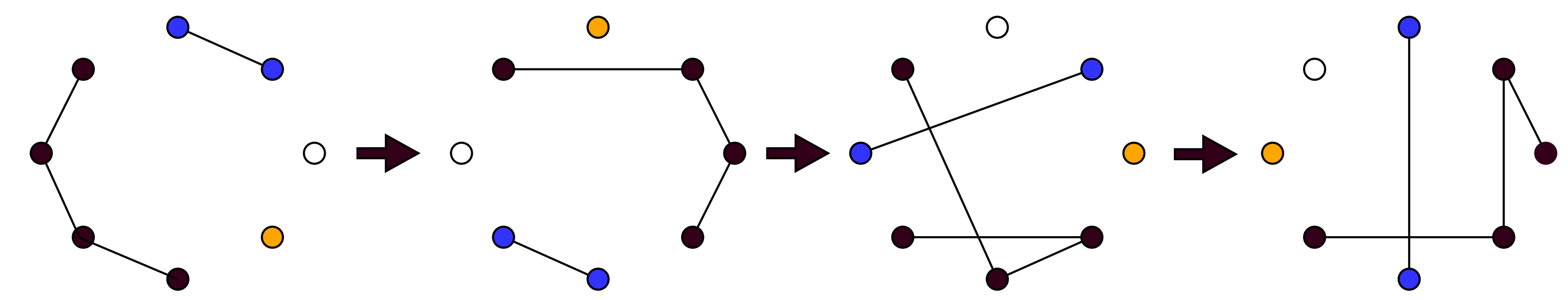
where $\bar{x}_{\text{active}}^{(t)} \coloneqq \frac{1}{k} \sum_{v_i \in V_{\text{active}}^{(t)}} x_i^{(t)}$ and $r_0 \coloneqq f(\bar{x}^{(0)}) - f^\star$.

Efficient Hyperparameter Search for k

- **Key idea:** If we utilize inactive nodes, we can run *TELEPORTATION* with various hyperparameter k in parallel.

Key Lemma

Let $k^\star \in \{1, 2, 3, \cdots, n\}$ be the optimal number of active nodes. If $k^\star < n$, there exists $k \in \{1, 2, 4, 8, \cdots, 2^{\lfloor \log_2(n+1) \rfloor - 1}\}$ that satisfies $\frac{k^\star}{4} < k \leq k^\star$. Furthermore, it holds that $\sum_{i=0}^{\lfloor \log_2(n+1) \rfloor - 1} 2^i \leq n$.



Algorithm 2 Efficient hyperparameter search for TELEPORTATION.

1: **Input:** the total number of iteration $2T$, and the number of nodes n.
2: run Alg. 1 for T iterations with number of active nodes n, and let $\{\{x_{n,i}^{(t)}\}_{i=1}^n\}_{t=0}^T$ denote the generated parameters.
3: **for** $k \in \{1, 2, 2^2, 2^3, \cdots, 2^{\lfloor \log_2(n+1) \rfloor - 1}\}$ **in parallel do**
4:   run Alg. 1 for T iterations with number of active nodes k.
5: **end for**
6: let $\{\{x_{k,i}^{(t)}\}_{v_i \in V_{\text{active}}^{(t)}}\}_{t=0}^T$ denote the parameters generated by Alg. 1 with k active nodes.
7: **return** the *best*[b] parameters among $k \in \{1, 2, 2^2, \cdots, 2^{\lfloor \log_2(n+1) \rfloor - 1}, n\}$.

[b]Theoretically, we select the parameters with the smallest gradient norm as in Theorem 2. In practice, we can select k that achieves the best accuracy on the validation datasets as in the vanilla grid search.

Thanks to the key lemma, Algorithm 2 can achieve the same rate as the rate shown in Theorem 1.

Experiments

- **Synthetic Function**

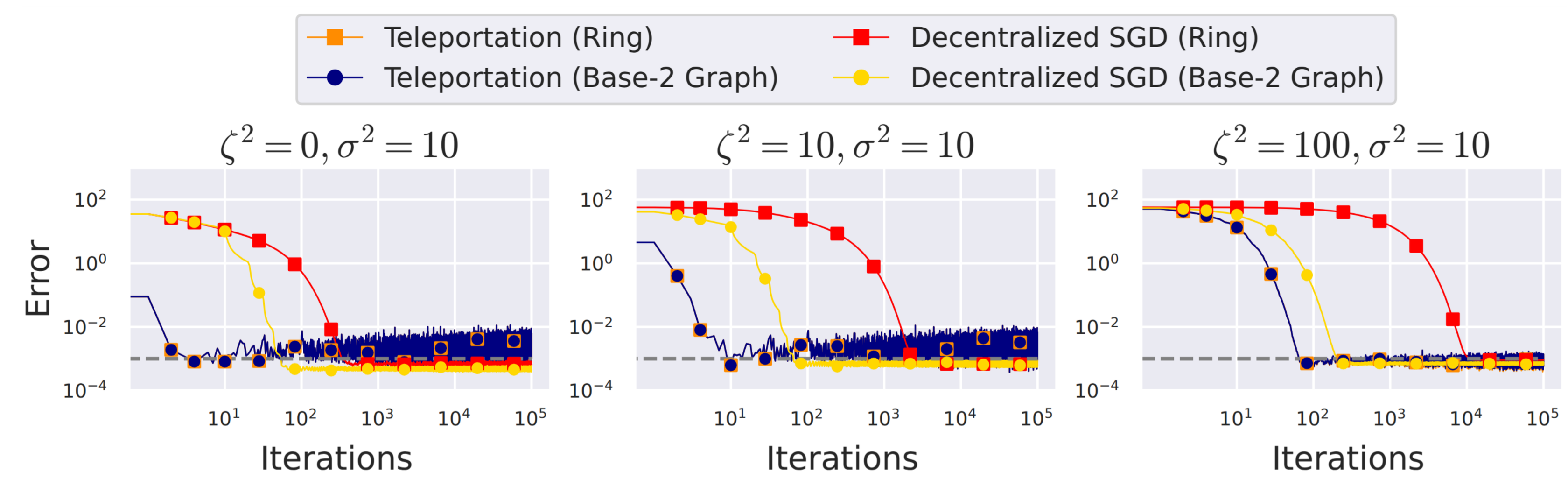


Figure: Convergence of the error to the target accuracy 0.001 when $n = 100$.

- **Neural Networks**

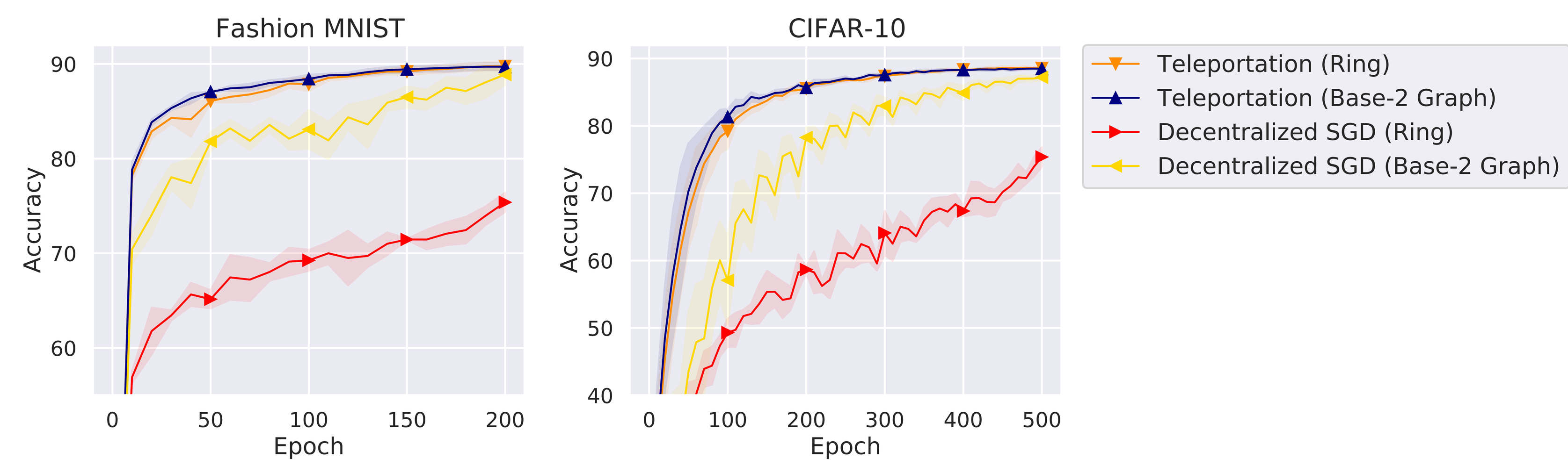


Figure: Test accuracy of TELEPORTATION and Decentralized SGD in a heterogeneous case.